

Enterprise Network Design using Cisco Packet Tracer

**Design and Implementation of a Multi-Zone
Enterprise Network using RIP, EIGRP,
OSPF, DHCP Relay, NAT, ACL, and GRE
VPN**

Submitted By

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Technologies Used

- Cisco Packet Tracer
- Cisco IOS
- IPv4 Networking
- RIP Version 2
- Enhanced Interior Gateway Routing Protocol (EIGRP)
- Open Shortest Path First (OSPF)
- Dynamic Host Configuration Protocol (DHCP Relay)
- Network Address Translation (NAT)
- Access Control Lists (ACL)
- Generic Routing Encapsulation (GRE Tunnel)
- Cisco 2811 Routers
- Cisco 2960 Switches

Project Type

Enterprise Campus Network Design

Certificate

Certificate of Completion

This is to certify that the project entitled "**Enterprise Network Design using Cisco Packet Tracer**" has been successfully designed and implemented by **Ramanand Patharwalkar** as a practical implementation of enterprise networking concepts.

The project demonstrates the implementation of modern networking technologies including multiple routing protocols, centralized IP address management, network security policies, secure tunneling, and enterprise infrastructure services. The design follows structured enterprise networking principles and reflects practical knowledge gained through academic study, networking laboratories, and Cisco CCNA certification.

This report documents the architecture, implementation methodology, configurations, testing procedures, and analysis of the complete enterprise network.

Acknowledgement

I would like to express my sincere gratitude to everyone who contributed directly or indirectly to the successful completion of this project.

I am thankful to my faculty members and mentors for providing the theoretical knowledge and practical guidance required to understand enterprise networking concepts. Their support enabled me to apply networking principles in a practical environment using Cisco Packet Tracer.

I would also like to acknowledge Cisco Networking Academy for providing excellent learning resources that strengthened my understanding of routing, switching, and enterprise network design.

Special thanks to the Centre for Development of Advanced Computing (CDAC) for enhancing my technical skills and encouraging practical implementation of networking technologies.

Finally, I would like to thank my family and friends for their continuous encouragement throughout the completion of this project.

Abstract

Modern organizations require secure, scalable, and highly available communication infrastructures capable of supporting thousands of users and multiple departments. Enterprise networks are therefore designed using hierarchical architectures that separate network responsibilities into multiple layers to improve scalability, performance, and security.

The objective of this project is to design and implement an enterprise campus network using Cisco Packet Tracer while demonstrating practical implementation of industry-standard networking technologies.

The network is divided into three major operational zones:

- Access Zone
- Core Operations Zone
- Distribution Zone

Each zone contains multiple departments connected through Cisco routers and Layer-2 switches.

The project implements three dynamic routing protocols:

- Routing Information Protocol (RIP)
- Enhanced Interior Gateway Routing Protocol (EIGRP)
- Open Shortest Path First (OSPF)

to simulate routing domains commonly found in enterprise environments.

To improve network functionality, several enterprise services have been integrated, including DHCP Relay for centralized IP address allocation, Static Network Address Translation (NAT) for private-to-public address mapping, Access Control Lists (ACLs) for traffic filtering, and a Generic Routing Encapsulation (GRE) Tunnel to simulate secure communication between remote sites.

The completed topology consists of twelve Cisco routers interconnected through multiple WAN links, seven Layer-2 switches, centralized servers, and several departmental LANs.

The project validates communication between all departments through extensive testing using ping, traceroute, routing table verification, DHCP address allocation, NAT translation, and ACL policy enforcement.

This implementation demonstrates the practical application of Cisco enterprise networking concepts and provides valuable experience in network planning, configuration, security, troubleshooting, and documentation.

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Introduction

Networking forms the foundation of modern information technology infrastructures. Every enterprise organization depends on a reliable communication network to facilitate data exchange, collaboration, centralized resource sharing, internet access, and secure communication among departments.

As organizations grow, the complexity of their networks increases significantly. Small office networks can no longer support the scalability, security, and performance requirements of medium and large enterprises. Consequently, organizations adopt hierarchical enterprise network architectures that divide responsibilities into different network layers.

The Cisco Enterprise Campus Architecture is one of the most widely adopted models for designing scalable and resilient networks. It divides network functions into Access, Distribution, and Core layers to simplify management while improving performance and availability.

This project implements a practical simulation of an enterprise campus network using Cisco Packet Tracer. The network integrates multiple routing technologies and enterprise services commonly deployed in production environments.

Unlike small laboratory topologies that use a single routing protocol, this project demonstrates coexistence of multiple routing protocols across different network zones. The implementation also incorporates centralized DHCP services, secure tunneling, network address translation, and traffic filtering mechanisms.

The resulting network closely resembles the architecture used in educational institutions, hospitals, financial organizations, software companies, manufacturing industries, and multinational corporations.

Problem Statement

Organizations often face several networking challenges as they expand.

Some of the common problems include:

- Poor scalability
- Inefficient routing
- Lack of centralized IP management
- Security vulnerabilities
- Network congestion
- Limited fault tolerance
- Complex administration
- Difficult troubleshooting

Small flat networks become increasingly difficult to manage because every device belongs to the same broadcast domain. Such networks suffer from excessive broadcast traffic, reduced performance, and limited scalability.

Furthermore, manually assigning IP addresses to every device increases administrative effort and creates opportunities for addressing conflicts.

Without proper routing protocols, communication between multiple departments becomes inefficient and difficult to maintain.

Similarly, unrestricted communication between departments introduces security risks that may expose confidential organizational data.

Therefore, enterprise organizations require structured network architectures capable of supporting secure, scalable, and efficient communication.

This project addresses these challenges by implementing:

- Hierarchical enterprise design
- Department segmentation
- Dynamic routing

- Centralized DHCP services
- Static NAT
- ACL-based traffic filtering
- GRE-based secure communication

Objectives

The primary objectives of this project are:

Primary Objectives

- Design a scalable enterprise network.
- Divide the organization into multiple operational zones.
- Connect all departments using Cisco routers.
- Implement dynamic routing protocols.
- Enable communication between all departments.
- Centralize IP address allocation using DHCP Relay.
- Implement Network Address Translation.
- Secure network communication using ACLs.
- Establish secure communication using GRE Tunnel.
- Validate complete network functionality.

Secondary Objectives

- Improve network scalability.
- Simplify network administration.
- Demonstrate enterprise routing concepts.
- Reduce manual IP configuration.
- Improve network security.
- Simulate real-world enterprise infrastructure.
- Enhance troubleshooting skills.
- Gain practical experience with Cisco IOS.
- Apply CCNA networking concepts.

Scope of the Project

The scope of this project includes the design, implementation, testing, and documentation of an enterprise network using Cisco Packet Tracer.

The project focuses on practical implementation of enterprise networking technologies rather than theoretical explanation alone.

The implementation includes:

- Enterprise network architecture
- Department-based segmentation
- Layer-2 switching
- Layer-3 routing
- Dynamic routing protocols
- DHCP Relay
- Static NAT
- ACL implementation
- GRE Tunnel
- Enterprise connectivity testing
- Network documentation

Although the project is developed using Cisco Packet Tracer, the networking concepts implemented are directly applicable to real Cisco enterprise networks.

Part 2 – Enterprise Network Architecture, Topology Design, Device Inventory & Communication Flow

Chapter 1

Enterprise Network Architecture

1.1 Introduction

Enterprise networks are designed to support communication between multiple departments while ensuring scalability, security, fault tolerance, and simplified administration. Unlike small office networks, enterprise infrastructures must efficiently manage hundreds or thousands of devices distributed across different organizational units.

To achieve these goals, enterprise networks are commonly built using a hierarchical architecture. This design divides the network into functional layers, allowing each layer to perform a dedicated role.

The network developed in this project follows a **three-zone enterprise architecture**, inspired by Cisco's hierarchical campus design principles. Instead of creating a flat network where every department is directly interconnected, the infrastructure is organized into logical zones that separate end-user access, core routing functions, and service distribution.

This modular approach improves performance, simplifies troubleshooting, and provides a scalable foundation for future expansion.

1.2 Enterprise Design Philosophy

The primary goal of this design is to create a network that satisfies the following requirements:

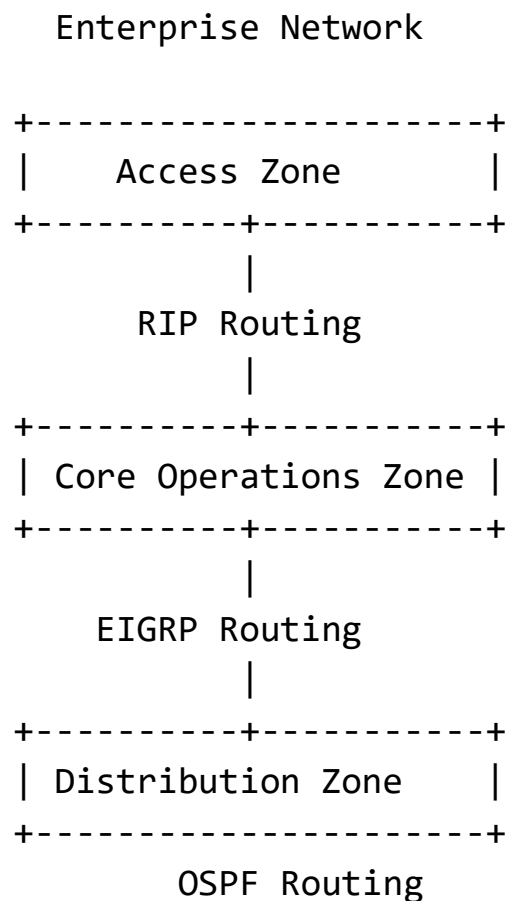
- Scalability
- High Availability
- Easy Administration

- Secure Communication
- Efficient Routing
- Centralized Services
- Department Isolation
- Future Expansion

Each department operates within its own IP subnet while remaining capable of communicating with other departments through dynamic routing protocols.

1.3 Three-Zone Enterprise Model

The implemented network consists of three major operational zones:



Each zone performs a unique function within the enterprise

Chapter 2

Access Zone Design

2.1 Overview

The Access Zone is responsible for connecting users to the enterprise network. It represents the first point of entry into the infrastructure and contains the routers and switches that provide connectivity to employee workstations.

This zone includes four departments:

- Human Resources
- Information Technology
- Training
- Development

Each department has:

- One Cisco Router
- One Layer-2 Switch
- Multiple End Devices
- Individual LAN
- Dedicated Gateway

The routers also function as default gateways for the departmental networks.

2.2 Responsibilities

The Access Zone performs several critical tasks:

- Connects end-user devices
- Provides gateway services
- Forwards packets toward the core
- Relays DHCP requests
- Participates in RIP routing
- Supports secure communication through GRE

2.3 Department Layout

Department	Router	LAN Network
HR	Router0	192.168.7.0/24
IT	Router2	192.168.6.0/24
Training	Router1	192.168.9.0/24
Development	Router3	192.168.8.0/24

2.4 Why RIP?

Routing Information Protocol (RIP) was selected for the Access Zone because:

- Small routing domain
- Simple configuration
- Automatic route advertisement
- Low processing overhead
- Suitable for departmental networks

Although RIP is not generally recommended for very large enterprise environments, it remains an excellent choice for small branch or departmental networks and demonstrates foundational dynamic routing concepts.

Chapter 3

Core Operations Zone

3.1 Overview

The Core Operations Zone serves as the backbone of the enterprise network.

All communication between departments ultimately passes through this layer before reaching other organizational units or external services.

Unlike the Access Zone, the Core Zone is optimized for high-speed packet forwarding rather than direct end-user connectivity.

3.2 Departments

The Core Zone consists of:

- Finance Department
- Research & Development
- Enterprise Services
- Data Center

3.3 Responsibilities

The Core performs the following functions:

- High-speed routing
- Traffic aggregation
- Inter-zone communication
- Enterprise backbone connectivity
- DHCP relay
- Static NAT

3.4 Why EIGRP?

Enhanced Interior Gateway Routing Protocol (EIGRP) provides several advantages over RIP:

- Fast convergence
- Low bandwidth consumption
- Low CPU utilization
- Support for unequal-cost load balancing
- Diffusing Update Algorithm (DUAL)

The Core Operations Zone requires rapid recovery from failures; therefore, EIGRP is an appropriate protocol for this layer.

Chapter 4

Distribution Zone

4.1 Overview

The Distribution Zone acts as the policy enforcement layer between the enterprise backbone and external services.

It aggregates traffic from multiple networks while implementing routing policies, security controls, and external connectivity.

Departments in this zone include:

- Sales & Marketing
- Supply Chain
- Remote Operations

4.2 Responsibilities

The Distribution Zone performs:

- Route summarization
- Policy enforcement
- NAT
- ACL implementation
- External connectivity
- OSPF routing

4.3 Why OSPF?

OSPF is widely used in enterprise environments because it offers:

- Link-state routing
- Fast convergence
- Hierarchical design
- Area support
- High scalability
- Efficient path calculation using SPF

These characteristics make OSPF ideal for distribution-layer deployments.

Chapter 5

Physical Topology

5.1 Overview

The physical topology consists of interconnected Cisco networking devices that emulate a real enterprise campus network.

Hardware Components

Device	Quantity
Cisco 2811 Routers	12
Cisco 2960 Switches	7
DHCP Servers	2
PCs	Multiple
Laptops	Multiple
WAN Serial Links	Multiple
Ethernet Links	Multiple

5.2 Router Connectivity

Each department is connected to neighboring routers using serial point-to-point links.

The routers exchange routing information through dynamic routing protocols.

The design provides multiple communication paths between zones, reducing dependency on a single route.

5.3 LAN Infrastructure

Every department contains:

PCs

|

Laptop

|

Switch

|

Router

This closely resembles enterprise office deployments.

Chapter 6

Logical Topology

Unlike the physical topology, the logical topology focuses on how data travels through the network.

Users



Department Router



Access Zone



Core Operations



Distribution



Destination Department

Each packet follows the routing tables learned through dynamic routing protocols rather than fixed paths.

Example Communication

HR PC



HR Router



Access Zone RIP



Core EIGRP



Distribution OSPF



Sales Router



Sales PC

This demonstrates inter-zone communication across different routing domains.

Chapter 7

Network Communication Flow

Communication follows these stages:

Step 1

A client generates data.

↓

Step 2

The packet reaches the departmental router.

↓

Step 3

The router consults its routing table.

↓

Step 4

Dynamic routing determines the optimal path.

↓

Step 5

The packet traverses the Core Zone.

↓

Step 6

The Distribution Zone applies routing and security policies.

↓

Step 7

The destination router forwards the packet to the target LAN.

Chapter 8

Device Inventory

Router	Department	Routing Protocol
Router0	HR	RIP
Router1	Training	RIP
Router2	IT	RIP
Router3	Development	RIP
Router4	Supply Chain	OSPF
Router5	Remote Operations	OSPF
Router6	Sales & Marketing	OSPF
Router7	Distribution	OSPF
Router8	Enterprise Services	EIGRP
Router9	R&D	EIGRP
Router10	Finance	EIGRP
Router11	Core Services	EIGRP

Layer 2 Switches

Cisco 2960 switches connect end-user devices to departmental routers.

Functions include:

- MAC address learning
- Frame forwarding
- Broadcast forwarding
- LAN segmentation

Servers

The enterprise contains centralized servers providing:

- DHCP Services
- Internal Resources
- Enterprise Applications

These servers are accessible from multiple departments using routed communication

Chapter 9

Enterprise Design Methodology

The network was developed using the following design methodology:

Phase 1

Requirement Analysis



Phase 2

Department Identification



Phase 3

IP Address Planning



Phase 4

Router Deployment



Phase 5

Switch Deployment



Phase 6

Routing Configuration



Phase 7

DHCP Configuration



Phase 8

ACL Implementation



Phase 9

NAT Configuration



Phase 10

GRE Tunnel Implementation



Phase 11

Testing & Validation

Part 3 – IP Addressing Scheme and Network Planning

Chapter 10

IP Addressing Scheme

10.1 Introduction

One of the most critical aspects of enterprise network design is the development of a structured IP addressing scheme. A well-planned addressing strategy improves scalability, simplifies troubleshooting, reduces configuration errors, and ensures efficient routing between departments.

In this project, a **private IPv4 addressing scheme** is used throughout the enterprise. Each department has its own dedicated subnet, while router-to-router WAN connections use separate networks to isolate traffic and simplify route management.

The addressing plan follows these design principles:

- Each department has a unique subnet.
- No overlapping IP ranges are used.
- Router interfaces act as default gateways.
- Centralized DHCP services assign IP addresses to end devices.
- Serial links use dedicated WAN subnets.
- Private IP addresses are translated using NAT where external communication is required.

10.2 Addressing Strategy

The enterprise network is divided into three logical addressing domains:

Zone	Purpose	Addressing Type
Access Zone	End-user departments	Private IPv4
Core Operations	Backbone communication	Private IPv4
Distribution Zone	Enterprise services and WAN	Private IPv4 + NAT

10.3 IPv4 Address Classes

The entire topology uses addresses from the **192.168.0.0/16** private network.

Advantages of using private IPv4 addressing include:

- No public IP requirement
- Increased security
- Suitable for internal communication
- Supports NAT implementation
- Scalable for enterprise environments

Chapter 11

Department LAN Addressing

Each department has its own LAN subnet with a dedicated default gateway configured on the router's FastEthernet interface.

Access Zone

Department	Network	Gateway
HR	192.168.7.0/24	192.168.7.1
IT	192.168.6.0/24	192.168.6.1
Training	192.168.9.0/24	192.168.9.1
Development	192.168.8.0/24	192.168.8.1

Core Operations Zone

Department	Network	Gateway
Enterprise Services	192.168.12.0/24	192.168.12.1
Finance	192.168.15.0/24	192.168.15.1
Core Services	192.168.17.0/24	192.168.17.1
R&D	192.168.20.0/24	192.168.20.1

Distribution Zone

Department	Network	Gateway
Supply Chain	192.168.23.0/24	192.168.23.1
Remote Operations	192.168.25.0/24	192.168.25.1
Sales & Marketing	192.168.27.0/24	192.168.27.1
Distribution Office	192.168.30.0/24	192.168.30.1

Chapter 12

WAN Addressing Scheme

The routers communicate through dedicated serial links. Each link is assigned a separate subnet to avoid conflicts and simplify routing.

Link	Network
HR ↔ Training	192.168.1.0/24
Training ↔ Development	192.168.2.0/24
Development ↔ IT	192.168.3.0/24
IT ↔ HR	192.168.4.0/24
IT ↔ DHCP Server	192.168.5.0/24
Development ↔ Enterprise Services	192.168.10.0/24
Training ↔ Supply Chain	192.168.11.0/24
Enterprise Services ↔ Finance	192.168.13.0/24
Enterprise Services ↔ R&D	192.168.14.0/24
Finance ↔ Core Services	192.168.16.0/24
Core Services ↔ R&D	192.168.19.0/24
R&D ↔ Sales	192.168.21.0/24
Supply Chain ↔ Remote Operations	192.168.22.0/24
Supply Chain ↔ Sales	192.168.24.0/24
Remote Operations ↔ Distribution	192.168.29.0/24
Sales ↔ Distribution	192.168.28.0/24

Chapter 13

Router Interface Mapping

Router 0 (HR)

Interface	Address	Purpose
Fa0/0	192.168.7.1	HR LAN
S0/2/0	192.168.4.2	IT Connection
S0/3/0	192.168.1.1	Training Connection

Router 1 (Training)

Interface	Address	Purpose
Fa0/0	192.168.6.1	IT LAN
Fa0/1	192.168.5.1	DHCP Server
S0/0/0	192.168.4.1	HR
S0/3/0	192.168.3.2	Development

Router 2 (IT)

Interface	Address	Purpose
Fa0/0	192.168.6.1	IT LAN
Fa0/1	192.168.5.1	DHCP Server
S0/0/0	192.168.4.1	HR
S0/3/0	192.168.3.2	Development

Router 3 (Development)

Interface	Address	Purpose
Fa0/0	192.168.8.1	Development LAN
S0/0/0	192.168.10.1	Enterprise Services
S0/2/0	192.168.3.1	IT
S0/3/0	192.168.2.2	Training

The remaining routers follow the same addressing pattern and are documented in the Router Configuration Analysis chapter.

Chapter 14

Default Gateway Planning

Each LAN uses the router's FastEthernet interface as the default gateway.

Department	Gateway
HR	192.168.7.1
IT	192.168.6.1
Training	192.168.9.1
Development	192.168.8.1
Finance	192.168.15.1
R&D	192.168.20.1
Sales	192.168.27.1

This approach provides a consistent and easily managed gateway assignment across the enterprise.

Chapter 15

DHCP Address Allocation

Rather than configuring static addresses for every client device, the project uses **centralized DHCP services**.

Department routers include the `ip helper-address` command, allowing DHCP broadcast requests to be forwarded to the central DHCP server.

Benefits

- Automatic IP assignment
- Centralized administration
- Reduced configuration effort
- Prevention of duplicate IP addresses
- Simplified onboarding of new devices

Chapter 16

Addressing Design Considerations

The IP addressing scheme was designed with the following considerations:

Scalability

Each department has a separate subnet, making it easy to add additional hosts without affecting other departments.

Security

Subnet separation limits broadcast domains and supports the implementation of ACLs to control inter-department communication.

Manageability

The use of structured addressing allows administrators to identify the department associated with an IP address quickly.

Routing Efficiency

Unique subnets simplify routing table entries and reduce the possibility of routing loops.

Future Expansion

Additional departments can be added by allocating unused private subnets without redesigning the existing network.

Chapter 17

Best Practices

Although /24 networks are used throughout the project for simplicity, production environments often optimize address usage by:

- Using /30 subnets for point-to-point WAN links.
- Reserving address ranges for infrastructure devices.
- Documenting DHCP exclusion ranges.
- Implementing route summarization where possible.
- Maintaining an IP Address Management (IPAM) system.

These practices improve scalability and operational efficiency in large enterprise networks.

Part 4 – Access Zone Design and Implementation

Chapter 18

Access Zone Design

18.1 Introduction

The **Access Zone** is the first layer of the enterprise network where end-user devices connect to the organizational infrastructure. It provides connectivity for employees, workstations, printers, laptops, and other endpoint devices while serving as the entry point into the enterprise backbone.

In this project, the Access Zone consists of four independent departments:

- Human Resources (HR)
- Information Technology (IT)
- Training
- Development

Each department has:

- One Cisco 2811 Router
- One Cisco 2960 Switch
- Multiple End Devices
- Independent LAN
- Dedicated Default Gateway

The Access Zone is configured using **Routing Information Protocol (RIP Version 2)** to enable dynamic route exchange

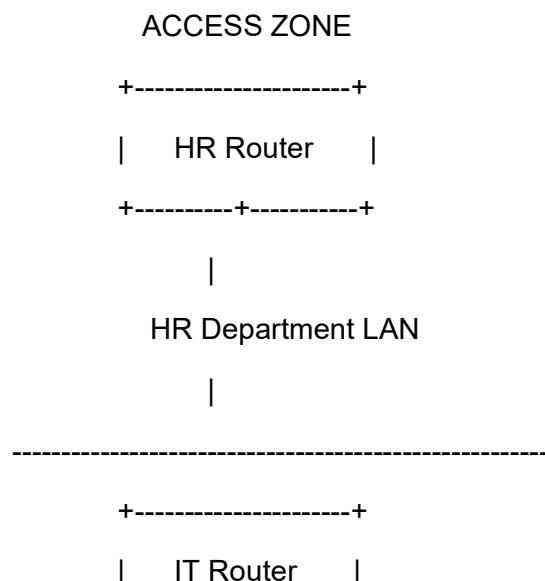
between departmental routers. DHCP Relay is implemented to provide centralized IP address assignment, while a GRE tunnel simulates secure communication between remote enterprise locations.

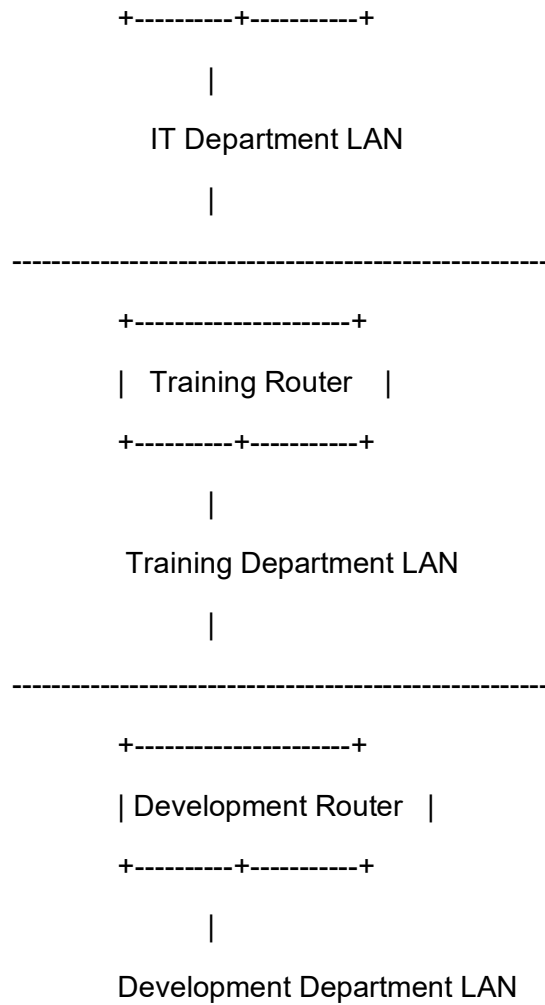
18.2 Design Objectives

The Access Zone was designed with the following objectives:

- Provide reliable connectivity to end-user devices.
- Enable dynamic routing between departments.
- Reduce manual IP configuration using DHCP Relay.
- Support secure communication with GRE.
- Ensure each department is isolated within its own subnet.
- Provide a scalable foundation for future expansion.

18.3 Access Zone Topology





Each router acts as the **default gateway** for its departmental subnet and exchanges routing information with neighboring routers using RIP.

Chapter 19

Human Resources Department (Router 0)

19.1 Overview

The Human Resources department is one of the four Access Zone departments. It connects HR employees to the enterprise network while maintaining a dedicated subnet.

Router0 performs three major functions:

- Default gateway for the HR LAN.
- RIP route advertisement.
- DHCP relay forwarding.

LAN

192.168.7.0/24

Gateway

192.168.7.1

19.2 Interface Configuration

Interface	Address	Function
FastEthernet0/0	192.168.7.1	HR LAN Gateway
Serial0/2/0	192.168.4.2	IT Connection
Serial0/3/0	192.168.1.1	Training Connection

These interfaces establish redundancy and allow packets to reach neighboring departments through dynamic routing.

19.3 DHCP Relay

Router0 contains the following command:

```
ip helper-address 192.168.5.2
```

Instead of broadcasting DHCP requests only within the HR LAN, the router forwards them to the centralized DHCP server.

Advantages

- Centralized administration
- Automatic address allocation
- Reduced maintenance
- Easier expansion

19.4 RIP Configuration

Router0 advertises:

```
192.168.1.0
```

```
192.168.4.0
```

```
192.168.7.0
```

These networks enable communication with:

- HR Department
- IT Department
- Training Department

Dynamic routing removes the need for manually configured static routes.

19.5 Packet Flow Example

HR PC



Switch



Router0



RIP



Core Network



Destination

Chapter 20

Training Department (Router 1)

20.1 Overview

The Training Department provides network connectivity for employee training laboratories and learning resources.

The router performs:

- Routing
- DHCP Relay
- GRE Tunnel endpoint

20.2 Interface Summary

Interface	Address	Purpose
Fa0/0	192.168.9.1	Training LAN
S0/3/0	192.168.1.2	HR
S0/2/0	192.168.2.1	Development
S0/1/0	192.168.11.1	Supply Chain

20.3 GRE Tunnel

One of the most significant features of Router1 is the GRE tunnel:

```
interface Tunnel0
```

```
ip address 192.168.100.1
```

```
tunnel source Serial0/2/0
```

```
tunnel destination 192.168.10.2
```

This tunnel establishes a logical connection with the Core Operations Zone.

Benefits

- Simulated VPN
- Encapsulation of routed traffic
- Secure communication
- Remote site connectivity

Although Packet Tracer does not implement full encryption, the GRE tunnel demonstrates enterprise WAN design concepts

20.4 RIP Networks

Router1 advertises:

- 192.168.1.0
- 192.168.2.0
- 192.168.9.0
- 192.168.11.0

These routes allow the Training Department to communicate with neighboring routers automatically.

Chapter 21

IT Department (Router 2)

21.1 Overview

The IT Department serves as one of the most critical departments in the Access Zone.

Besides acting as a gateway for the IT LAN, it connects to the DHCP server network and implements an Access Control List (ACL).

21.2 Interface Summary

Interface	Address	Function
Fa0/0	192.168.6.1	IT LAN
Fa0/1	192.168.5.1	DHCP Server
S0/0/0	192.168.4.1	HR
S0/3/0	192.168.3.2	Development

21.3 DHCP Server Connectivity

Unlike the other departmental routers, Router2 directly connects to the subnet containing the centralized DHCP server.

This allows DHCP relay requests from other departments to reach the DHCP service.

21.4 ACL Configuration

Router2 implements an Extended ACL:

```
access-list 101 deny ip host 192.168.5.2 192.168.26.0 0.0.0.255
```

```
access-list 101 permit ip any any
```


Policy Explanation

The ACL prevents the DHCP server host (192.168.5.2) from directly communicating with the Remote Operations subnet (192.168.26.0/24), while permitting all other traffic. This demonstrates how extended ACLs can enforce inter-department security policies.

21.5 Security Advantages

The ACL provides:

- Department isolation
- Unauthorized access prevention
- Controlled communication
- Improved network security

Chapter 22

Development Department (Router 3)

22.1 Overview

The Development Department provides connectivity for software development systems. The router connects the department with:

- IT
- Training
- Enterprise Services

making it one of the key transit points within the Access Zone.

22.2 Interface Summary

Interface	Address	Purpose
Fa0/0	192.168.8.1	Development LAN
S0/2/0	192.168.3.1	IT
S0/3/0	192.168.2.2	Training
S0/0/0	192.168.10.1	Core Operations

22.3 RIP Networks

Router3 advertises:

- 192.168.2.0
- 192.168.3.0
- 192.168.8.0
- 192.168.10.0

This configuration enables the Development Department to exchange routing information dynamically with neighboring routers and provides a bridge toward the Core Operations Zone.

Chapter 23

Routing within the Access Zone

The four Access Zone routers exchange routing information using RIP Version 2.

Route Advertisement

Each router advertises:

- Local LAN subnet
- WAN serial links
- Neighbor connections

This enables all departments to learn remote networks dynamically without the need for static routes.

Example Route Learning :

Development Router

↓

Advertises 192.168.8.0

↓

Training Router

↓

HR Router

↓

IT Router

Chapter 24

Packet Flow Through the Access Zone

The following illustrates a typical communication path from an HR workstation to a Development workstation:

HR PC

↓

HR Switch

↓

Router 0

↓

RIP

↓

Router 1

↓

Router 3

↓

Development Switch

↓

Development PC

At each hop, the router consults its routing table, determines the best next hop, and forwards the packet accordingly.

Chapter 25

Design Evaluation

The Access Zone design offers several benefits:

- **Scalability:** Each department has its own subnet and router, allowing new departments to be added with minimal impact.
- **Centralized Services:** DHCP relay reduces administrative overhead by avoiding separate DHCP servers for each LAN.
- **Security:** ACLs restrict unauthorized communication between selected networks.
- **Dynamic Routing:** RIP automatically adapts to topology changes without manual intervention.
- **Remote Connectivity:** The GRE tunnel demonstrates how branch locations can be connected over routed networks.

While RIP is appropriate for this educational project, larger production networks would typically migrate to OSPF or EIGRP within the access layer to improve convergence and scalability.

Part 5 – Core Operations

Zone Design and Implementation

Chapter 26

Core Operations Zone

26.1 Introduction

The **Core Operations Zone** is the backbone of the enterprise network. It is responsible for transporting data between different departments with high speed, reliability, and minimal latency. Unlike the Access Zone, where users connect to the network, the Core Zone focuses on efficient packet forwarding and inter-zone communication.

In this project, the Core Operations Zone connects the Access Zone with the Distribution Zone and hosts critical enterprise services such as Finance, Research & Development (R&D), Enterprise Services, and Core Services.

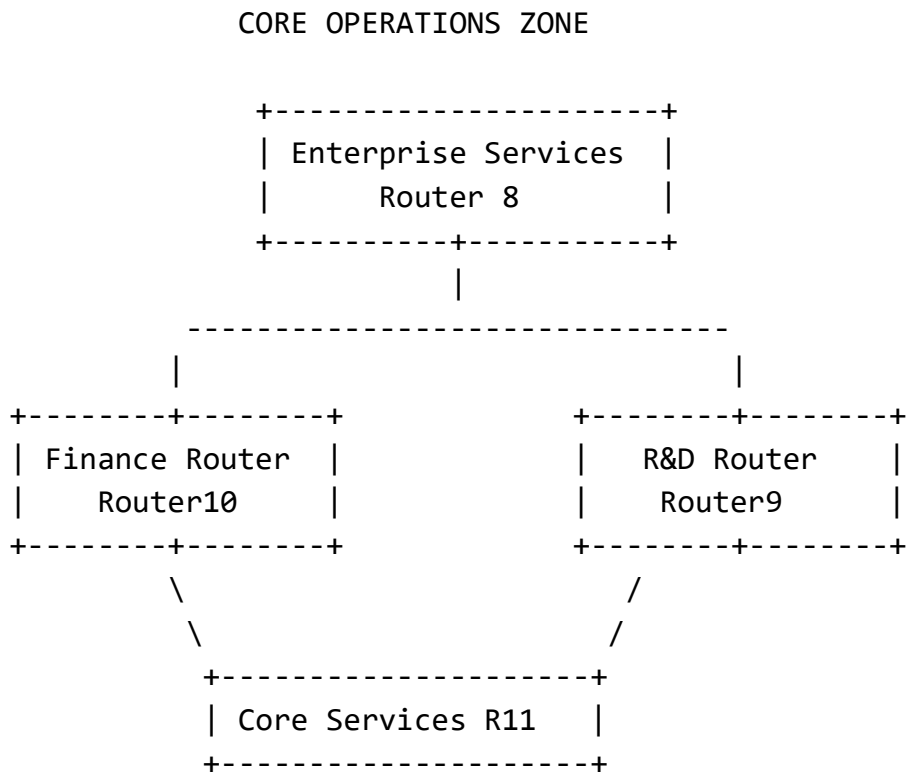
This zone uses **Enhanced Interior Gateway Routing Protocol (EIGRP)** due to its rapid convergence, efficient bandwidth utilization, and scalability.

26.2 Design Objectives

The Core Operations Zone was designed to:

- Provide high-speed routing between network zones.
- Act as the enterprise backbone.
- Support scalable routing with EIGRP.
- Connect enterprise service networks.
- Provide centralized connectivity to distribution networks.
- Support DHCP relay and NAT services.
- Enable secure communication through GRE tunnels.

26.3 Core Zone Architecture



The topology provides multiple routing paths and ensures reliable communication between departments.

Chapter 27

Enterprise Services Router (Router 8)

27.1 Overview

Router 8 functions as the **Enterprise Services Router**, connecting the Access Zone to the Core Operations Zone. It serves as the primary gateway for enterprise services and participates in EIGRP routing.

Responsibilities

- Connects Development Department to Core.
- Exchanges EIGRP routes.
- Participates in the GRE tunnel.
- Routes traffic to Finance and R&D.
- Supports centralized services.

27.2 Interface Configuration

Interface	IP Address	Connected To
Fa0/0	192.168.12.1	Enterprise Services LAN
S0/0/0	192.168.10.2	Development Router
S0/1/0	192.168.13.1	Finance Router
S0/2/0	192.168.14.1	R&D Router
Tunnel0	192.168.100.2	Training Router

The router acts as the aggregation point for traffic entering the Core Zone from the Development Department.

27.3 EIGRP Configuration

Router 8 advertises the following networks through EIGRP:

- 192.168.10.0/24
- 192.168.12.0/24
- 192.168.13.0/24
- 192.168.14.0/24

Why EIGRP?

EIGRP was selected because it:

- Converges quickly after failures.
- Uses the DUAL algorithm to prevent routing loops.
- Supports unequal-cost load balancing.
- Generates partial updates, reducing bandwidth consumption.

27.4 GRE Tunnel Termination

Router 8 is the destination endpoint of the GRE tunnel initiated from Router 1.

Tunnel Configuration

```
interface Tunnel0
 ip address 192.168.100.2 255.255.255.0
 tunnel source Serial0/0/0
 tunnel destination 192.168.2.1
```

This tunnel simulates secure site-to-site communication between the Access Zone and the Core Zone, encapsulating traffic within a virtual point-to-point link.

Chapter 28

Finance Department (Router 10)

28.1 Overview

The Finance Department hosts critical business applications and financial resources. Due to the sensitivity of financial data, this department relies on secure and efficient routing.

Responsibilities

- Default gateway for Finance LAN.
- EIGRP route advertisement.
- Connectivity to Enterprise Services and Core Services.

28.2 Interface Summary

Interface	Address	Purpose
Fa0/0	192.168.15.1	Finance LAN
S0/0/0	192.168.13.2	Enterprise Services
S0/1/0	192.168.16.1	Core Services

28.3 EIGRP Networks

The router advertises:

- 192.168.13.0
- 192.168.15.0
- 192.168.16.0

This enables Finance resources to be reachable from all other enterprise departments.

28.4 Design Considerations

Financial systems require:

- Reliable connectivity.
- Fast convergence during failures.
- Controlled access.
- Low latency.

EIGRP meets these requirements through rapid route recalculation and efficient bandwidth utilization.

Chapter 29

Research & Development Department (Router 9)

29.1 Overview

The R&D Department supports engineering and innovation teams. It communicates with Enterprise Services and the Distribution Zone.

Responsibilities

- Gateway for R&D subnet.
- EIGRP routing.
- Transit routing toward Sales.

29.2 Interface Summary

Interface	Address	Connected To
Fa0/0	192.168.20.1	R&D LAN
S0/0/0	192.168.14.2	Enterprise Services
S0/1/0	192.168.19.2	Core Services
S0/2/0	192.168.21.1	Sales Router

29.3 Role in Enterprise Communication

Router 9 bridges the Core and Distribution Zones. Traffic destined for the Sales Department traverses this router before entering the OSPF domain.

Chapter 30

Core Services Router (Router 11)

30.1 Overview

Router 11 provides connectivity for Core Services and hosts Network Address Translation (NAT) functionality.

Responsibilities

- Core Services gateway.
- Static NAT configuration.
- EIGRP participation.
- Connectivity to Finance and R&D.

30.2 Interface Summary

Interface	Address	Purpose
Fa0/0	192.168.17.1	Core Services LAN
S0/0/0	192.168.16.2	Finance
S0/1/0	192.168.19.1	R&D

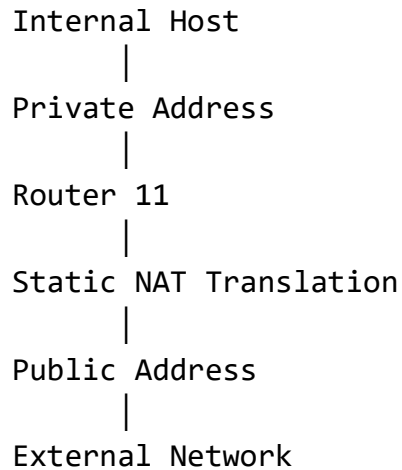
30.3 Static NAT

Router 11 implements Static NAT to translate private enterprise addresses into externally reachable addresses.

Purpose of NAT

- Hide internal addressing.
- Allow controlled external access.
- Conserve public IP addresses.
- Improve security

NAT Workflow



This configuration demonstrates enterprise edge routing practices where internal resources are selectively exposed through address translation.

Chapter 31

EIGRP Operation

The Core Zone routers exchange routing information using **Enhanced Interior Gateway Routing Protocol (EIGRP)**.

EIGRP Features

- Classless routing.
- Fast convergence.
- DUAL algorithm.
- Incremental updates.
- Support for VLSM.
- Loop-free routing.

Neighbor Discovery

Each router establishes EIGRP neighbor relationships over directly connected serial links. Once neighbors are discovered, routing information is exchanged automatically.

Chapter 32

Inter-Zone Communication

The Core Operations Zone acts as the central transit layer.

Example communication:

1. HR PC sends a packet.
2. Router0 forwards it using RIP.
3. Development Router forwards it to Router8.
4. Router8 routes it through EIGRP.
5. Finance Router delivers the packet to the destination LAN.

This layered forwarding model minimizes routing complexity and supports scalable expansion.

Chapter 33

Advantages of the Core Design

The implemented Core Zone provides several enterprise benefits:

High Availability

Multiple router interconnections reduce dependency on a single communication path.

Scalability

Additional departments can be integrated into the Core without redesigning the entire network.

Performance

EIGRP minimizes convergence time, reducing service interruptions during topology changes.

Centralized Services

Enterprise applications, DHCP relay, and NAT are efficiently supported from the Core.

Simplified Troubleshooting

The modular architecture isolates faults to specific network zones, making diagnostics more straightforward.

Chapter 34

Core Zone Evaluation

The Core Operations Zone successfully demonstrates:

- Enterprise backbone routing.
- Dynamic route exchange with EIGRP.
- GRE tunnel integration.
- Static NAT implementation.
- Efficient inter-zone packet forwarding.
- High-speed communication between departments.

This design reflects many principles used in real-world enterprise campus networks, including hierarchical routing, modular architecture, and centralized services.

Part 6 – Distribution Zone Design and Implementation

Chapter 35

Distribution Zone

35.1 Introduction

The **Distribution Zone** serves as the policy enforcement and service aggregation layer of the enterprise network. It connects the Core Operations Zone to enterprise branches and remote departments while applying routing policies, security controls, and network services.

In this project, the Distribution Zone consists of four routers:

- Router 4 – Supply Chain
- Router 5 – Remote Operations
- Router 6 – Sales & Marketing
- Router 7 – Distribution Router

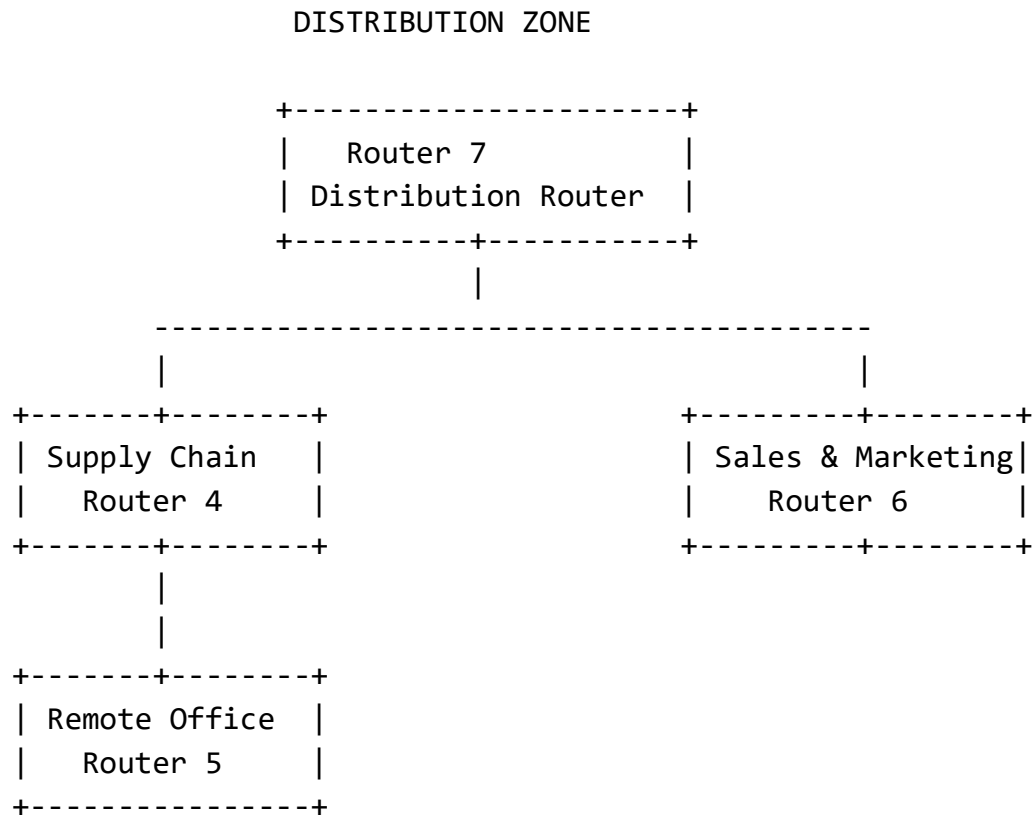
These routers exchange routing information using **Open Shortest Path First (OSPF)**, a link-state routing protocol widely adopted in enterprise environments due to its scalability, rapid convergence, and support for hierarchical network design.

35.2 Objectives

The Distribution Zone was designed to achieve the following objectives:

- Aggregate traffic from multiple enterprise departments.
- Connect remote locations to the enterprise backbone.
- Implement scalable dynamic routing using OSPF.
- Enforce security policies through ACLs.
- Provide Network Address Translation (NAT) for external communication.
- Act as the transition point between the Core and remote networks.

35.3 Distribution Zone Architecture



This design provides multiple communication paths while allowing centralized policy enforcement.

Chapter 36

Supply Chain Department (Router 4)

36.1 Overview

The Supply Chain Department manages logistics and inventory operations. Router 4 connects the Supply Chain LAN to the Distribution Zone and exchanges routes with neighboring routers using OSPF.

Responsibilities

- Default gateway for Supply Chain users.
- OSPF route advertisement.
- Connectivity to Training, Sales, and Remote Operations.
- Traffic forwarding toward the Core Zone.

36.2 Interface Summary

Interface	Address	Connected To
FastEthernet0/0	192.168.23.1	Supply Chain LAN
Serial0/0/0	192.168.11.2	Training Router
Serial0/1/0	192.168.22.1	Remote Operations
Serial0/2/0	192.168.24.1	Sales Router

36.3 OSPF Configuration

Router 4 advertises:

- 192.168.11.0/24
- 192.168.22.0/24
- 192.168.23.0/24
- 192.168.24.0/24

These advertisements allow neighboring routers to build a complete link-state database and calculate optimal routes using the Shortest Path First (SPF) algorithm.

Chapter 37

Remote Operations Department (Router 5)

37.1 Overview

Router 5 represents a remote branch office connected to the enterprise network. It provides LAN connectivity for remote users and implements Static NAT to simulate communication with external networks.

Responsibilities

- Gateway for Remote Operations LAN.
- OSPF participation.
- Static NAT.
- ACL enforcement.
- Connectivity to Distribution Router.

37.2 Interface Summary

Interface	Address	Purpose
FastEthernet0/0	192.168.25.1	Remote Operations LAN
Serial0/0/0	192.168.22.2	Supply Chain
Serial0/1/0	192.168.29.1	Distribution Router

37.3 Static NAT

Router 5 performs Static NAT to translate selected internal addresses into externally reachable addresses.

NAT Process

Private Host
↓
Router 5
↓
Static NAT Translation
↓
Public Address
↓
External Network

This allows internal systems to communicate with external resources while preserving the privacy of the internal addressing scheme.

Benefits

- Hides internal IP addresses.
- Conserves public IP space.
- Controls exposure of internal services.
- Improves security.

37.4 ACL Implementation

Router 5 includes an extended ACL to restrict communication between selected hosts and networks.

Example policy:

```
access-list 100 deny ip host 192.168.30.2 192.168.5.0 0.0.0.255  
access-list 100 permit ip any any
```

This rule prevents traffic from a specific host in the Distribution subnet from reaching the DHCP server network while permitting all other traffic. Such selective filtering reflects common enterprise security practices.

Chapter 38

Sales & Marketing Department (Router 6)

38.1 Overview

The Sales & Marketing Department connects business users to enterprise services and external resources. Router 6 serves as the gateway for the department and participates in OSPF.

Responsibilities

- Gateway for Sales LAN.
- OSPF routing.
- Connectivity to R&D.
- Connectivity to Distribution Router.
- Connectivity to Supply Chain.

38.2 Interface Summary

Interface	Address	Connected To
FastEthernet0/0	192.168.27.1	Sales LAN
Serial0/0/0	192.168.21.2	R&D
Serial0/1/0	192.168.24.2	Supply Chain
Serial0/2/0	192.168.28.1	Distribution Router

38.3 OSPF Route Advertisement

Router 6 advertises:

- 192.168.21.0/24
- 192.168.24.0/24
- 192.168.27.0/24
- 192.168.28.0/24

These routes enable dynamic communication between the Distribution and Core Zones.

Chapter 39

Distribution Router (Router 7)

39.1 Overview

Router 7 functions as the aggregation point within the Distribution Zone. It collects routes from neighboring OSPF routers and forwards traffic toward remote departments.

Responsibilities

- OSPF route aggregation.
- Traffic forwarding.
- Connectivity to Remote Operations.
- Connectivity to Sales.
- Distribution LAN gateway.

39.2 Interface Summary

Interface	Address	Purpose
FastEthernet0/0	192.168.30.1	Distribution LAN
Serial0/0/0	192.168.28.2	Sales
Serial0/1/0	192.168.29.2	Remote Operations

39.3 Role in Enterprise Design

Router 7 ensures that traffic entering the Distribution Zone is directed toward the appropriate departmental subnet while maintaining routing information through OSPF.

Chapter 40

OSPF Operation

The Distribution Zone uses **Open Shortest Path First (OSPF)** as its Interior Gateway Protocol.

Why OSPF?

OSPF offers several advantages:

- Link-state routing.
- Fast convergence.
- Efficient bandwidth utilization.
- Classless routing support.
- Hierarchical design using areas.
- Loop-free path selection.

Although this project uses a single OSPF area, the design can be extended into multiple areas for larger enterprise deployments.

OSPF Neighbor Formation

Each router establishes OSPF neighbor relationships over directly connected serial links. Once adjacencies are formed, routers exchange Link-State Advertisements (LSAs), build a synchronized Link-State Database (LSDB), and independently calculate the shortest paths using the SPF algorithm.

Chapter 41

Packet Flow through the Distribution Zone

Example communication from the Finance Department to a Remote Operations workstation:

Finance PC

↓

Router10 (EIGRP)

↓

Router9

↓

Router6 (OSPF)

↓

Router7

↓

Router5

↓

Remote Operations PC

The transition from EIGRP in the Core Zone to OSPF in the Distribution Zone demonstrates communication across multiple routing domains within the enterprise.

Chapter 42

Security within the Distribution Zone

Security is implemented through a combination of:

- Extended Access Control Lists (ACLs).
- Static NAT.
- Departmental subnet isolation.
- Controlled routing.

These mechanisms ensure that only authorized traffic reaches sensitive enterprise resources while preserving connectivity for legitimate users.

Chapter 43

Distribution Zone Evaluation

The Distribution Zone successfully demonstrates several enterprise networking principles:

Scalability

OSPF allows additional routers and departments to be integrated with minimal configuration changes.

Policy Enforcement

ACLs provide granular traffic filtering based on source, destination, protocol, and network.

Efficient Routing

The link-state approach enables routers to calculate optimal paths quickly after topology changes.

External Connectivity

Static NAT provides controlled communication between private enterprise networks and external systems.

Reliability

Multiple serial links create redundant communication paths, improving overall network resilience.

Part 7 – Enterprise Services, Testing, Troubleshooting, Best Practices, Future Scope & Conclusion

Chapter 44

Enterprise Services Integration

44.1 Introduction

A modern enterprise network is more than a collection of routers and switches. It integrates centralized services that simplify administration, improve scalability, enhance security, and provide reliable communication between departments.

In this project, the following enterprise services were implemented:

- Dynamic Routing
- DHCP Relay
- Static Network Address Translation (NAT)
- Access Control Lists (ACLs)
- GRE Tunnel (VPN Simulation)

Together, these technologies provide a secure and scalable networking environment suitable for medium-sized enterprise deployments.

Chapter 45

Routing Protocol Analysis

One of the key strengths of this project is the implementation of **multiple routing protocols** within a single enterprise topology. Each protocol was selected based on the role of the corresponding network zone.

45.1 RIP (Routing Information Protocol)

Purpose

RIP was deployed in the **Access Zone** to provide simple and efficient routing between departmental routers.

Characteristics

- Distance Vector Protocol
- Hop Count Metric
- Maximum 15 Hops
- Periodic Updates Every 30 Seconds
- Easy Configuration
- Low Processing Overhead

Advantages

- Simple to configure.
- Suitable for small routing domains.
- Ideal for educational environments.
- Minimal hardware requirements.

Limitations

- Slow convergence.
- Limited scalability.
- Hop count restriction.
- Less efficient than EIGRP and OSPF.

45.2 EIGRP (Enhanced Interior Gateway Routing Protocol)

Purpose

EIGRP was selected for the **Core Operations Zone**, where rapid convergence and efficient routing are essential.

Characteristics

- Cisco Proprietary (historically)
- DUAL Algorithm
- Composite Metric
- Partial Updates
- Fast Convergence

Advantages

- Excellent scalability.
- Low bandwidth consumption.
- Rapid failure recovery.
- Supports unequal-cost load balancing.

45.3 OSPF (Open Shortest Path First)

Purpose

OSPF was implemented in the **Distribution Zone**, where policy enforcement and scalability are critical.

Characteristics

- Link-State Routing Protocol
- SPF Algorithm
- Hierarchical Design
- Classless Routing
- Fast Convergence

Advantages

- Enterprise standard.
- Supports very large networks.
- Efficient routing updates.
- Excellent scalability.

45.4 Protocol Comparison

Feature	RIP	EIGRP	OSPF
Type	Distance Vector	Advanced Distance Vector	Link State
Metric	Hop Count	Bandwidth & Delay	Cost
Convergence	Slow	Fast	Fast
Scalability	Low	High	Very High
Classless Support	Yes (RIPv2)	Yes	Yes
Enterprise Usage	Small Networks	Medium to Large	Large Enterprises

Chapter 46

DHCP Relay

46.1 Overview

Instead of deploying individual DHCP servers in each department, the project uses a centralized DHCP server. Departmental routers relay DHCP requests using the `ip helper-address` command.

DHCP Process

Client

↓

DHCP Discover

↓

Router

↓

IP Helper Address

↓

DHCP Server

↓

DHCP Offer

↓

Router

↓

Client

Advantages

- Centralized administration.
- Automatic IP address assignment.
- Reduced configuration effort.
- Easier maintenance.
- Improved scalability.

Chapter 47

Network Address Translation (NAT)

47.1 Purpose

The project uses Static NAT on edge routers to enable communication between private enterprise networks and external networks.

NAT Workflow

Private Host

↓

Router

↓

Private IP

↓

NAT Translation

↓

Public IP

↓

External Network

Benefits

- Conserves public IP addresses.
- Hides internal network structure.
- Improves security.
- Allows controlled external access.

Chapter 48

Access Control Lists (ACLs)

48.1 Purpose

ACLs provide packet filtering by allowing or denying traffic based on source address, destination address, protocol, or port.

Types Implemented

- Standard ACL
- Extended ACL

Example Use Cases

- Restrict access between departments.

- Protect critical enterprise services.
- Control communication with server networks.
- Prevent unauthorized access.

Security Benefits

- Network segmentation.
- Reduced attack surface.
- Controlled resource access.
- Policy enforcement.

Chapter 49

GRE Tunnel

49.1 Overview

A Generic Routing Encapsulation (GRE) Tunnel was configured between the Training Router and the Enterprise Services Router to simulate secure site-to-site communication.

Purpose

- Logical point-to-point connection.
- Encapsulation of routed traffic.
- Simulated VPN connectivity.

Enterprise Applications

- Branch office connectivity.
- WAN communication.
- Inter-site routing.
- Overlay networks.

Chapter 50

Testing and Validation

After completing the network configuration, several tests were performed to verify functionality.

50.1 Connectivity Testing

Ping

Verified communication between:

- HR ↔ IT
- HR ↔ Finance
- Finance ↔ Sales
- Development ↔ Remote Office
- Supply Chain ↔ Training

Expected Result: Successful replies from all reachable departments.

50.2 Traceroute

Traceroute was used to confirm that packets traversed the expected sequence of routers between different zones.

50.3 Routing Verification

The following commands were used:

```
show ip route
show ip protocols
show ip eigrp neighbors
show ip ospf neighbor
show ip rip database
```

These commands verified that routing tables and neighbor relationships were correctly established.

50.4 DHCP Verification

Client devices were checked to ensure they received:

- Valid IP address.
- Correct subnet mask.
- Default gateway.
- DNS server (if configured).

50.5 NAT Verification

The following command confirmed active NAT translations:

```
show ip nat translations
```

50.6 ACL Verification

ACLs were tested by attempting communication that should be:

- Permitted.
- Denied.

This confirmed that security policies were functioning as intended.

50.7 GRE Tunnel Verification

The tunnel status and connectivity were verified using:

```
show interfaces tunnel0  
ping <Tunnel IP>
```

Successful replies confirmed that the GRE tunnel was operational.

Chapter 51

Troubleshooting Guide

Common Issues

Routing Problems

Symptoms

- Destination unreachable.
- Missing routes.
- Incomplete routing tables.

Verification

```
show ip route  
show ip protocols
```

Resolution

- Verify advertised networks.
- Check interface status.
- Confirm neighbor relationships.

DHCP Issues

Symptoms

- Clients receive APIPA addresses.
- No IP assignment.

Verification

`show running-config`

Check:

- `ip helper-address`
- DHCP pool configuration
- Interface status

NAT Issues

Symptoms

- Internal devices cannot access external resources.

Verification

`show ip nat translations`

`show running-config`

Ensure:

- Correct inside/outside interfaces.
- Proper static mappings.

OSPF Problems

Symptoms

- Missing routes.
- Neighbor not forming.

Verify

```
show ip ospf neighbor
```

EIGRP Problems

Verify

```
show ip eigrp neighbors
```

Check:

- Autonomous System Number.
- Network statements.
- Interface status.

Chapter 52

Best Practices

To improve maintainability and align with enterprise standards, the following practices are recommended:

- Use descriptive router hostnames (e.g., HR-RTR, CORE-RTR, SALES-RTR).
- Replace /24 point-to-point WAN links with /30 or /31 subnets.
- Disable EIGRP auto-summary (no auto-summary).
- Implement route redistribution instead of running RIP across multiple domains.
- Configure VLANs and inter-VLAN routing.
- Add HSRP or VRRP for gateway redundancy.
- Enable Cisco Express Forwarding (CEF).
- Configure SNMP, Syslog, and NTP for monitoring.
- Document all ACLs with clear naming conventions.
- Use secure management protocols such as SSH instead of Telnet.

Chapter 53

Future Enhancements

The current design provides a strong enterprise foundation and can be extended with advanced networking features.

Planned Enhancements

- VLAN implementation.
- Inter-VLAN routing.
- IPv6 addressing.
- Dynamic NAT with PAT.
- HSRP gateway redundancy.
- EtherChannel.
- Spanning Tree optimization.
- Wireless LAN Controller integration.
- Active Directory and DNS.
- AAA Authentication (RADIUS/TACACS+).
- QoS for voice and video traffic.
- SNMP monitoring.
- Syslog server.
- Firewall integration (Cisco ASA or Firepower).
- Network automation using Ansible or Python.

Chapter 54

Skills Demonstrated

This project demonstrates proficiency in:

- Enterprise Network Design
- Cisco IOS Configuration
- IPv4 Addressing

- RIP Configuration
- EIGRP Configuration
- OSPF Configuration
- DHCP Relay
- Static NAT
- Access Control Lists
- GRE Tunnel Configuration
- Layer 2 Switching
- Layer 3 Routing
- Network Security
- Enterprise Troubleshooting
- Technical Documentation

Chapter 55

Conclusion

This project successfully demonstrates the design and implementation of a **multi-zone enterprise network** using Cisco Packet Tracer. By dividing the topology into Access, Core Operations, and Distribution Zones, the network reflects a hierarchical architecture commonly used in enterprise environments.

The implementation integrates multiple dynamic routing protocols—RIP, EIGRP, and OSPF—to showcase different routing strategies within the same infrastructure. Centralized DHCP relay simplifies IP address management, while Static NAT enables communication between private and external networks. Access Control Lists enforce security policies, and the GRE tunnel simulates secure site-to-site connectivity.

Through comprehensive testing, including connectivity verification, routing validation, NAT inspection, ACL policy checks, and GRE tunnel verification, the project demonstrates reliable end-to-end communication and correct operation of all configured services.

This project strengthened practical skills in network planning, routing, switching, security, troubleshooting, and documentation. It also provides a strong portfolio example that aligns with real-world enterprise networking practices and showcases readiness for entry-level Network Engineer, NOC Engineer, or Infrastructure Engineer roles.

